Jacobian Analysis of a Recurrent Transformer Block

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A recurrent block is a discrete dynamical system

$$Y^{(t+1)} = F_\nu(Y^{(t)}), \qquad Y^{(t)} \in \mathbb{R}^{S \times D}.$$

- The same block parameters are reused at every iteration.
- The one-step Jacobian measures how an input perturbation changes after one update.
- We bound this Jacobian for a block containing attention, an FFN, residual connections, and RMSNorm.

Tomihari and Karakida (2025) give the corresponding normalization argument for recurrent self-attention. Here it is composed through both sub-blocks.

The update under study

$$
\begin{aligned}
Z_1 &= Y + \mathrm{MSA}(Y), & Y' &= R_1(Z_1),\\
Z_2 &= Y' + \mathrm{FFN}(Y'), & \mathrm{TB}(Y) &= R_2(Z_2),\\
\Delta(Y) &= C + \mathrm{TB}(Y), & U_\nu &= Y + \nu\Delta(Y),\\
& & F_\nu(Y) &= R_U(U_\nu).
\end{aligned}
$$

- R_1, R_2, R_U are separate, row-wise normalization layers.
- The context C is fixed during each trajectory and when computing $J_{F_\nu}(Y)$.
- In the trajectory experiment, $C = 0.1\,Y^{(0)}$.

RMSNorm: convention and derivative

For a token $y \in \mathbb{R}^D$ and layer $\ell \in \{1, 2, U\}$,

$$\rho_\ell(y) = \sqrt{\|y\|_2^2 / D + \epsilon_\ell}, \qquad R_\ell(y) = \mathrm{diag}(\gamma_\ell) \frac{y}{\rho_\ell(y)}, \quad \epsilon_\ell > 0.$$

$$J_{R_\ell}(y) = \frac{\mathrm{diag}(\gamma_\ell)}{\rho_\ell(y)} \left(I - \frac{y y^\top}{D \rho_\ell(y)^2} \right), \qquad \|J_{R_\ell}(y)\|_2 \leq \frac{g_\ell}{\rho_\ell(y)},$$

where $g_\ell = \|\gamma_\ell\|_\infty$.

- The factor decreases as the normalization input grows.
- It can exceed one when the input denominator is small.
- With $\epsilon_\ell > 0$, the matrix in parentheses is not an exact projector.

Theorem 3.1: one-step Jacobian bound

Use the smallest denominator across tokens at each layer:

$$r_1 = \min_i \rho_1((Z_1)_i), \quad r_2 = \min_i \rho_2((Z_2)_i), \quad r_U = \min_i \rho_U((U_\nu)_i).$$

Let $M = \|J_{\mathrm{MSA}}(Y)\|_2$ and $L = \|J_{\mathrm{FFN}}(Y')\|_2$. Then

$$\boxed{\|J_{F_\nu}(Y)\|_2 \le \frac{g_U}{r_U}\left[1 + \nu\frac{g_2 g_1}{r_2 r_1}(1+L)(1+M)\right].}$$

- This follows from the chain rule and submultiplicativity.
- Component norms are evaluated at their actual inputs.
- An average token denominator cannot replace the minimum.

Theorem 3.2: boundedness as the step grows

Fix Y, C, all weights, and all normalization parameters. Suppose

$$\delta = \min_i \|\Delta(Y)_i\|_2 > 0.$$

Let $B = \max_i \|Y_i\|_2$ and $K = g_2 g_1 (1 + L)(1 + M)/(r_2 r_1)$. For $\nu \geq 2B/\delta$,

$$r_U \geq \frac{\nu\delta}{2\sqrt{D}}, \qquad \|J_{F_\nu}(Y)\|_2 \leq \frac{2\sqrt{D}\, g_U}{\delta}\left(K + \frac{1}{\nu}\right) = O(1).$$

The outer denominator grows with ν and compensates for the step multiplier. The result is pointwise in the fixed state and parameters.

Bounded in step size does not mean convergent in time

For fixed context, derivatives through a trajectory involve a product:

$$\frac{\partial Y^{(T)}}{\partial Y^{(0)}} = J_{F_\nu}(Y^{(T-1)}) \cdots J_{F_\nu}(Y^{(0)}).$$

- A constant one-step bound $k > 1$ allows the product bound k^T.
- Local asymptotic stability of a fixed point follows from spectral radius less than one; a norm below one is a stronger sufficient condition.
- A large one-step singular value alone does not prove chaos or exponential growth along a trajectory.

For example, $x \mapsto -x$ on a sphere has bounded derivative and period-two orbits. This illustrates why a bounded norm is insufficient.

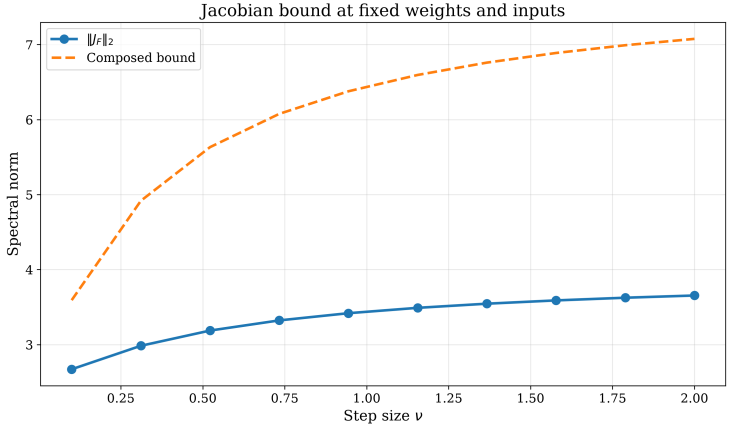
Experimental protocol

All blocks and image projections are untrained. Normalization scales start at one and $\epsilon = 10^{-8}$.

Experiment	S	D	H	D_{ff}
One-step bound	16	32	4	64
Large-step sweep	16	32	4	64
Trajectories	32	64	4	128

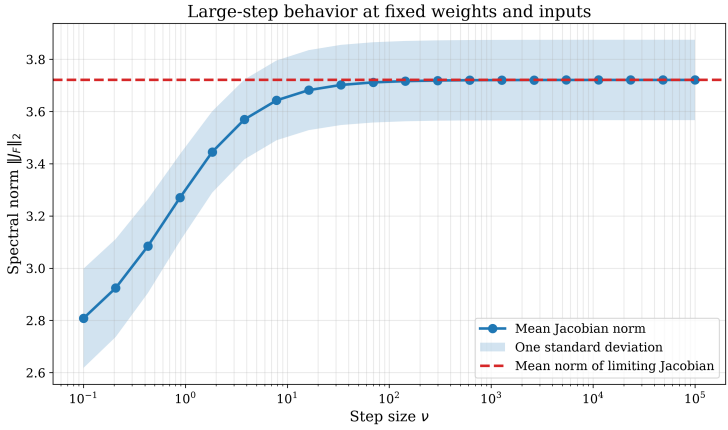
- Experiments 1 and 2: dense autograd Jacobians and SVD, float64.
- Weights, state, and context are fixed within each step-size sweep.
- Trajectories: 1000 steps, $\nu \in \{0.1, 0.5, 1.0\}$; 25 trials per dataset and step size.

Experiment 1: check the composed bound



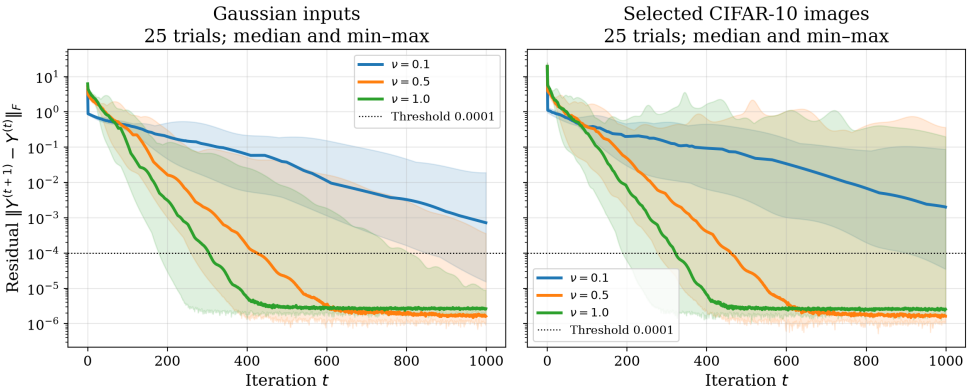
Seed 42; ten step sizes in $[0.1, 2.0]$. Both component Jacobian norms and all three token minima enter the bound.

Experiment 2: compare fixed-model sweeps



Twenty step sizes from 10^{-1} to 10^{5}; seeds 0–9. Shading is one standard deviation across seeds. At $\nu = 10^{5}$, the mean norm is 3.720649; the independently computed limiting mean is 3.720655.

Experiment 3: measure successive-state differences



Seeds 0–4; five Gaussian inputs and CIFAR-10 test images 0–4 per seed. Curves show medians; bands span the observed minimum and maximum.

Interpreting the trajectory experiment

The criterion is $d_t = \left\| Y^{(t+1)} - Y^{(t)} \right\|_F \leq 10^{-4}$ throughout the final 20 of 1000 steps.

Input	$\nu = 0.1$	$\nu = 0.5$	$\nu = 1.0$
Gaussian	6/25	23/25	25/25
CIFAR-10	6/25	20/25	22/25

- The counts increase with ν in this sample; this is not a general guarantee.
- Missing the threshold by step 1000 does not prove non-convergence.
- The residual measures progress toward a fixed point on one trajectory, not contraction between different initial states.

What the analysis establishes

- An explicit bound for the one-step Jacobian of the complete update.
- Separate contributions from attention, FFN, and each normalization layer.
- A bounded large-step limit at fixed state and parameters when every update row is nonzero.

Contraction and gradient control over many recurrent steps require additional conditions. They are not consequences of the large-step bound alone.

References and materials

A. Tomihari and R. Karakida (2025). *Recurrent Self-Attention Dynamics: An Energy-Agnostic Perspective from Jacobians*. arXiv:2505.19458v3. Proposition 5.1 and Lemma A.3.

B. Zhang and R. Sennrich (2019). *Root Mean Square Layer Normalization*. NeurIPS 32.

Code, revised manuscript, and numerical records

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